

EXPERTISE — what clinicians know about these columns

The scorer's model is fixed: a default `LogisticRegression()`. It draws one straight line through whatever numbers you give it. It cannot decide that an age of 999 is not an age, that a diagnosis is not a scale, or that an empty albumin cell means "the lab was not ordered". You decide that, and the reasons are written down: SUPPORT was a prognostic study, and its investigators documented how each value was collected, what a normal value is, and what they did when a value was missing.

This brief collects the twelve facts that matter most for preprocessing, with the choice each one justifies. It does not rank them: measuring them is your job. Column meanings are in `DICTIONARY.pdf`.

1. The cohort and the outcome

SUPPORT (Study to Understand Prognoses and Preferences for Outcomes and Risks of Treatments) followed 9,105 seriously ill adults admitted to five US academic medical centres between 1989 and 1994. A patient entered the study on the hospital day when one of the qualifying diagnoses was met (fact 6). The physiology you receive is the **worst value recorded on the third day after study entry**, as the investigators collected it for their prognostic model. The label is the patient's vital status two months (60 days) after entry: `dead = 1` if the patient died within that window.

What it justifies. Every column is something known at the bedside on day 3, with one exception (fact 10). Each row is one patient; the site is a hospital, not a quantity.

Source: Knaus et al., *Ann Intern Med* 1995; hbiostat.org/data/repo/supportdesc.

2. A lab that was not ordered is presumed normal

A test is ordered when the physician wants its result. When none was ordered, the investigators did not treat the value as unknown: they filled it with a normal value, and they published the values they used.

lab	SUPPORT fill-in	unit
albumin (alb)	3.5	g/dL
PaO2/FiO2 (pafi)	333.3	mmHg
bilirubin (bili)	1.01	mg/dL
creatinine (crea)	1.01	mg/dL
blood urea nitrogen (bun)	6.51	mg/dL
white blood cells (wblc)	9	$\times 10^9/L$
urine output (urine)	2502	mL/day

Arterial pH and glucose are not in their list. The textbook normals are pH 7.40 (reference range 7.35 to 7.45) and glucose about 100 mg/dL (fasting 70 to 100).

The median of the cohort is not a normal value, because the cohort is sick: in the source file the median albumin is 2.9 g/dL (normal 3.5 to 5.0), the median BUN is 23 mg/dL (normal 7 to 20), the median PaO2/FiO2 is 224 (above 400 on room air in a healthy adult).

What it justifies. Filling an empty lab with the cohort's median asserts that the untested patient looks like the tested ones. The investigators' own choice was the opposite. The table gives the domain fill, the median gives the generic one; which to use is yours to measure.

Source: hbiostat.org/data/repo/supportdesc, "Baseline physiologic variables"; Harrison's Principles of Internal Medicine, appendix of laboratory values.

3. Which absences carry information

Not every empty cell is the same kind of empty.

- **The patient could not answer.** `adlp` is the number of activities of daily living the *patient* said they could not do. It is empty when the patient could not be interviewed on day 3: intubated, comatose, delirious. `adls` is the same count reported by a *surrogate* (usually a family member), which is why it is filled far more often. The SUPPORT team built a single ADL score by taking the patient's answer when it existed and the surrogate's otherwise.
- **The arterial blood gas.** `pafi` and `ph` come from the same arterial sample, drawn when there is a respiratory or acid-base concern. They are missing together, and the absence means no such concern was raised on day 3.
- **The liver panel.** `alb` and `bili` are ordered as one panel.
- **The metabolic work-up.** `glucose`, `bun` and the 24-hour `urine` collection are missing together, as a set.
- **The interview.** `edu` and `income` are filled only when the social interview took place.

What it justifies. For each column, decide whether empty means "unknown", "not ordered because nothing suggested it", or "the patient could not speak", and whether the fact of the absence deserves a column of its own.

Source: hbiostat.org/data/repo/supportdesc (`adlp`, `adls`, `adlsc`); Knaus et al. 1995.

4. Real extremes versus impossible values

Seriously ill patients have extreme values, and most extremes in this file are real. A few are not measurements at all. The reference range and the physiology tell them apart.

column	normal (adult)	extreme but real	not a measurement
<code>age</code>	18 and over in this cohort	100	999 (placeholder for "unknown")
<code>temp</code>	36.5 to 37.5 °C	31.7 °C (severe hypothermia); 41.7 °C	0.0
<code>hrt</code>	60 to 100 /min	0 (arrest); 300 (a flutter conducted 1:1 is about the ceiling)	well above 300
<code>meanbp</code>	70 to 100 mmHg	0 (circulatory arrest); 195 (hypertensive crisis)	
<code>wblc</code>	4 to 11 ×10 ⁹ /L	0 (marrow aplasia after chemotherapy); 200 (leukaemia)	
<code>bili</code>	0.1 to 1.2 mg/dL	60 (fulminant liver failure)	
<code>crea</code>	0.6 to 1.2 mg/dL	20 (untreated end-stage kidney failure, dialysis level)	

column	normal (adult)	extreme but real	not a measurement
glucose	70 to 100 mg/dL fasting	1000 (hyperosmolar hyperglycaemic state)	
sod	135 to 145 mEq/L	about 110 to 180 is the range seen in the living	1370 (a decimal slip for 137.0)
urine	800 to 2000 mL/day	0 (anuria); 9000 (polyuria)	

What it justifies. A clipping rule written without the physiology removes real patients: the zero mean arterial pressure and the white cell count of 200 belong to the sickest people in the cohort. A placeholder left in place is a different error: one 999 in `age` is a lever on a straight line.

Source: *Harrison's Principles of Internal Medicine, appendix; MSD Manual Professional Edition, "Normal laboratory values"*.

5. Sites chart in different units

The five hospitals chart in their own conventions.

- **Creatinine:** sites A, B and C report mg/dL; **sites D and E report $\mu\text{mol/L}$.** 1 mg/dL = 88.4 $\mu\text{mol/L}$. A normal creatinine reads 1.0 at site A and 88 at site D.
- **Temperature:** sites A, B, D and E chart $^{\circ}\text{C}$; **site C charts $^{\circ}\text{F}$.** $^{\circ}\text{F} = ^{\circ}\text{C} \times 9/5 + 32$. A fever of 38.5 $^{\circ}\text{C}$ reads 101.3 at site C.

What it justifies. Any threshold, ratio or comparison across sites is meaningless until the units agree; the SIRS cut-offs and the BUN/creatinine ratio of fact 7 assume $^{\circ}\text{C}$ and mg/dL. Harmonise first, then look for extremes: a creatinine of 700 is an impossible mg/dL value and an ordinary $\mu\text{mol/L}$ one.

Source: *site conventions in DICTIONARY.pdf; conversion factors, MSD Manual*.

6. Diagnoses are categories, severities are orders, sex and race are neither

`dzgroup` is the qualifying diagnosis, in eight groups: ARF/MOSF w/Sepsis (acute respiratory failure, or multiple organ system failure with sepsis); MOSF w/Malig (multiple organ system failure in a patient with a malignancy); Coma (non-traumatic); CHF (congestive heart failure); COPD (chronic obstructive pulmonary disease, acute exacerbation); Cirrhosis (with a decompensation); Colon Cancer (with metastases); Lung Cancer (non-small-cell, advanced stage).

These are not points on one scale: a coma and a lung cancer kill through different mechanisms and at different speeds. `dzclass` groups the same column into four: ARF/MOSF = {ARF/MOSF w/Sepsis, MOSF w/Malig}; COPD/CHF/Cirrhosis = {COPD, CHF, Cirrhosis}; Cancer = {Colon Cancer, Lung Cancer}; Coma = {Coma}. It contains nothing that `dzgroup` does not.

Two columns are orders. `ca` is cancer status: none < non-metastatic < metastatic. `income` has four brackets: under \$11k < \$11-\$25k < \$25-\$50k < over \$50k. `sex` and `race` are nominal, and for a prognostic score they are not risk factors in themselves: the SUPPORT model did not use them.

What it justifies. A diagnosis needs an encoding that gives each group its own effect. A severity can keep its order as one number, or be split into categories; both cost different numbers of columns. Encoding a nominal column as an order (male = 1, female = 2) asserts a direction that does not exist.

Source: *Knaus et al. 1995 (entry criteria, model variables); hbiostat supportdesc*.

7. What clinicians compute at the bedside

Each of these is a formula on columns you have, with a published threshold.

- **SIRS** (ACCP/SCCM 1992): one point each for temperature > 38 or < 36 °C, heart rate > 90 /min, respiratory rate > 20 /min, white cells > 12 or $< 4 \times 10^9$ /L. Two or more points define SIRS; the count runs 0 to 4.
- **Shock index** = heart rate / systolic pressure (normal 0.5 to 0.7). Systolic pressure is not in this file; its variant, the **modified shock index** = heart rate / MAP, is (Liu et al. 2012: normal about 0.7 to 1.3, mortality rises above 1.3).
- **BUN/creatinine ratio**: above 20 suggests prerenal azotaemia (the kidney is under-perfused rather than damaged). Both values in mg/dL.
- **Berlin definition of ARDS** by $\text{PaO}_2/\text{FiO}_2$: mild 200 to 300, moderate 100 to 200, severe below 100; above 300 is not ARDS.
- **Oliguria**: fewer than 500 mL of urine in 24 hours; anuria fewer than 100 (KDIGO 2012 states its criterion per hour, < 0.5 mL/kg/h, which for an adult lands in the same region).
- **Hypoalbuminaemia**: albumin below 3.5 g/dL (below 2.5 is severe). **Acidaemia**: pH below 7.35; alkalaemia above 7.45. **Dysnatraemia**: sodium outside 135 to 145 mEq/L, in either direction.
- **The ADL count** (`adlp`, `adls`): 0 to 7 basic self-care activities the patient cannot do unaided (bathing, dressing, eating, getting out of bed, walking). Higher is more dependent.
- **The SUPPORT coma score** (`scoma`): the Glasgow Coma Scale rescaled to 0 (fully conscious) to 100 (deep coma).

Many labs are **log-normal**: bilirubin, creatinine, BUN and white cells span two orders of magnitude, and clinicians think in doublings, not differences (KDIGO stages kidney injury by the *ratio* of creatinine to its baseline: 1.5 $\times$, 2 $\times$, 3 $\times$). A white cell count is abnormal in both directions (fact 4).

What it justifies. A straight line cannot compute a ratio, a threshold or a logarithm from raw columns. Each formula above is a candidate feature with a clinical reason behind it; which ones pay, on this cohort, is what you measure.

Source: Bone et al., *Chest* 1992; Liu et al., *World J Emerg Med* 2012; ARDS Definition Task Force, *JAMA* 2012; KDIGO 2012; Knaus et al. 1995.

8. Scales differ by five orders of magnitude

Hospital charges are in dollars (thousands to over a million), urine output in mL (0 to 9,000), $\text{PaO}_2/\text{FiO}_2$ in the hundreds, glucose up to 1,100, creatinine in $\mu\text{mol/L}$ in the hundreds at two sites. Arterial pH lives between 6.8 and 7.8, albumin between 0.4 and 5.

What it justifies. The scorer's solver (`lbfgs`) takes steps that the widest-scaled column dominates, and it stops after 100 iterations whether or not it has arrived; its L2 penalty also weighs a coefficient by the unit its column happens to be in. Scaling is a decision about the solver, not about medicine, and a scaled placeholder is still a placeholder.

Source: *scikit-learn* documentation, `LogisticRegression (lbfgs, max_iter=100, C=1.0)`.

9. The same information twice

- `dzclass` is a function of `dzgroup` (fact 6).
- `bun` and `crea` both measure kidney function and rise together; their *ratio* is its own signal (fact 7).

- `adlp` and `adls` measure the same dependence from two reporters.

What it justifies. Two copies of one signal do not give a regularised line more to work with: the penalty splits one coefficient between them, and the fit gets less stable. Where two columns say the same thing, keep the better one, or a combination that says something new.

Source: hbiostat.org/data/repo/supportdesc; Knaus et al. 1995.

10. Information that exists only after the outcome

`charges` is the total hospital bill for the stay. It is computed at discharge, and it grows with the length of the stay and the intensity of care, so it also reflects how the stay ended. For a patient still in the ward there is no bill yet. In `train.csv` the column is filled for 98 % of the patients; in `test.csv` it is empty for every patient.

What it justifies. A model cannot use at prediction time what does not exist at prediction time. A column that is strong on the training rows and absent on the test rows makes the train AUC rise and the test AUC fall; the scorer reports both, and warns when they part.

Source: hbiostat.org/data/repo/supportdesc (`charges`); Knaus et al. 1995 (the model uses day-3 data only).

11. Age, comorbidities and the day of entry

`age` is the age at study entry. `num_co` counts the comorbid conditions recorded on entry (0 to 9); `diabetes` and `dementia` are two of them singled out. `hday` is the hospital day on which the patient met the entry criteria: most patients qualified on the day of admission, a minority after weeks in hospital, up to 148 days.

What it justifies. Age and the comorbidity count are already numbers on sensible scales. `hday` is not: the step from day 1 to day 3 does not mean what the step from day 101 to day 103 means, and a few long stays sit far from the rest.

Source: hbiostat.org/data/repo/supportdesc; Knaus et al. 1995.

12. The official model is a regression on these very columns

The SUPPORT prognostic model estimated each patient's probability of surviving two and six months from the qualifying diagnosis, age, the number of days in hospital before entry, the coma score, the comorbidities and the day-3 physiology. It was a Cox proportional-hazards regression (the survival cousin of a logistic regression) with the transformation of each variable chosen by the statisticians together with the clinicians. Sex and race were not inputs.

What it justifies. The fixed logistic regression of this challenge is the professionals' tool on this problem, given the professionals' variables. The features they gave it were constructed, not raw; the raw columns are what the chart contained.

Source: Knaus et al., *Ann Intern Med* 1995;122:191-203.

Sources

- Knaus WA, Harrell FE, Lynn J, et al. *The SUPPORT prognostic model: objective estimates of survival for seriously ill hospitalized adults*. Ann Intern Med 1995;122:191-203.
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- Harrison's Principles of Internal Medicine, 21st ed. (2022), appendix "Laboratory values of clinical importance"; MSD Manual Professional Edition, *Normal laboratory values*: <https://www.msdmanuals.com/professional/resources/normal-laboratory-values>
- SUPPORT2 on the UCI repository: <https://archive.ics.uci.edu/dataset/880/support2>